

```

1  clc;
2  clear;
3  close all;
4
5  % Input combinations
6  inputs = [0 0;
7            0 1;
8            1 0;
9            1 1];
10
11  disp('=====');
12  disp(' McCulloch-Pitts Model Logic Gates');
13  disp('=====');
14

```

```

15  %% AND Gate
16  disp(' ');
17  disp('AND Gate');
18
19  w = [1 1];
20  theta = 2;
21
22  for i = 1:4
23      net = inputs(i,:) * w';
24      if net >= theta
25          y = 1;
26      else
27          y = 0;
28      end
29      fprintf('%d %d -> %d\n', inputs(i,1), inputs(i,2), y);
30  end
31

```

```

32  %% OR Gate
33  disp(' ');
34  disp('OR Gate');
35
36  w = [1 1];
37  theta = 1;
38
39  for i = 1:4
40      net = inputs(i,:) * w';
41      if net >= theta
42          y = 1;
43      else
44          y = 0;
45      end
46      fprintf('%d %d -> %d\n', inputs(i,1), inputs(i,2), y);
47  end
48

```

```

49 %% NOT Gate
50 disp(' ');
51 disp('NOT Gate');
52
53 input_not = [0;1];
54 w = -1;
55 theta = 0;
56
57 for i = 1:2
58     net = input_not(i) * w;
59     if net >= theta
60         y = 1;
61     else
62         y = 0;
63     end
64     fprintf('%d -> %d\n', input_not(i), y);
65 end
66

```

```

67 %% XOR Gate (2-Layer McCulloch-Pitts Network)
68 disp(' ');
69 disp('XOR Gate');
70
71 for i = 1:4
72
73     A = inputs(i,1);
74     B = inputs(i,2);
75
76     % Hidden Layer
77     OR_out = (A + B) >= 1;    % OR neuron
78     AND_out = (A + B) >= 2;    % AND neuron
79
80     % Output Layer
81     % XOR = OR AND NOT(AND)
82     if (OR_out == 1 && AND_out == 0)
83         XOR_out = 1;
84     else
85         XOR_out = 0;
86     end
87
88     fprintf('%d %d -> %d\n', A, B, XOR_out);
89
90 end

```

McCulloch-Pitts Model Logic Gates

AND Gate

0 0 -> 0

0 1 -> 0

1 0 -> 0

1 1 -> 1

OR Gate

0 0 -> 0

0 1 -> 1

1 0 -> 1

1 1 -> 1

NOT Gate

0 -> 1

1 -> 0

XOR Gate

0 0 -> 0

0 1 -> 1

1 0 -> 1

1 1 -> 0

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